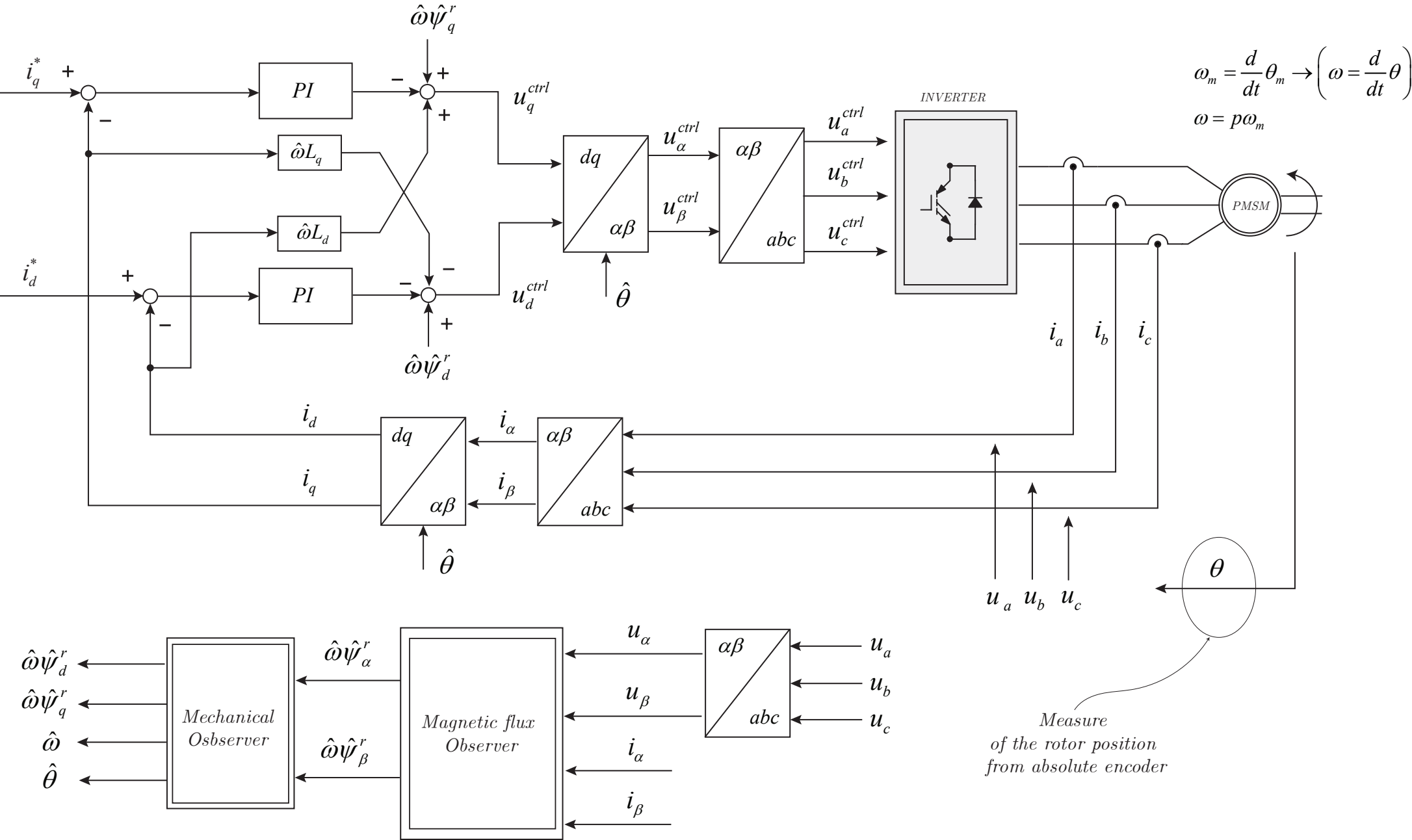

$\omega_m = \frac{d}{dt}\theta_m \rightarrow \left(\omega = \frac{d}{dt}\theta\right)$
$\omega = p\omega_m$
i_q^*
i_d^*
PI
PI
$\hat{\omega}L_q$
$\hat{\omega}L_d$
$\hat{\omega}\hat{\psi}_q^r$
$\hat{\omega}\hat{\psi}_d^r$
u_q^{ctrl}
u_d^{ctrl}
dq
αβ
$\hat{\theta}$
u_α^{ctrl}
u_β^{ctrl}
αβ
abc
u_a^{ctrl}
u_b^{ctrl}
u_c^{ctrl}
INVERTER
PMSM
i_a
i_b
i_c
i_d
i_q
dq
αβ
$\hat{\theta}$
i_α
i_β
αβ
abc
u_a
u_b
u_c
θ
Measure
of the rotor position
from absolute encoder
$\hat{\omega}\hat{\psi}_d^r$
$\hat{\omega}\hat{\psi}_q^r$
$\hat{\omega}$
$\hat{\theta}$
Mechanical
Osbserver
$\hat{\omega}\hat{\psi}_\alpha^r$
$\hat{\omega}\hat{\psi}_\beta^r$
Magnetic flux
Observer
u_α
u_β
i_α
i_β
αβ
abc
u_a
u_b
u_c